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A Wearable IMU System for Real-Time Skiing Posture Monitoring and Motion Feedback

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***Abstract* - This paper presents a five-node wearable IMU system for real-time alpine skiing posture monitoring and interpretable motion feedback. The nodes are mounted on the left boot cuff, right boot cuff, left thigh, right thigh, and chest, and transmit roll/pitch estimates to an ESP32 gateway through ESP-NOW. The gateway supports both USB streaming to a computer dashboard and Wi-Fi hotspot streaming to a smartphone application. Instead of full-body motion capture or direct skill scoring, the system focuses on key-segment feedback for common beginner errors. A lower-limb alignment indicator, a backseat tendency indicator, and reference-based dynamic curves are introduced for posture and motion analysis. Controlled angle validation achieved an average MAE of 0.6767° and RMSE of 0.8119°. Static and treadmill skiing experiments further showed selective responses to A-frame, backseat-like posture, upper-body engagement, and dynamic coordination differences. These results demonstrate the feasibility of distributed wearable IMUs for portable skiing posture monitoring and preliminary motion quality feedback.**

Index Terms* - *wearable sensing, inertial measurement unit, skiing posture, motion feedback, motion quality assessment

I. Introduction

Alpine skiing requires coordinated control of the skis, lower limbs, trunk, and fore-aft posture. Beginner and intermediate skiers often have difficulty perceiving posture errors in real time, leading to problems such as A-frame posture, lower-limb misalignment, backseat posture, weak-side turns, and abrupt transitions. Although coaches can identify these problems visually, such feedback is usually subjective, delayed, and difficult to quantify continuously.

Video analysis and optical motion capture can provide objective motion information, but they are limited by viewpoint, occlusion, cost, and fixed infrastructure. Wearable IMUs provide a portable sensing solution for field-based sport performance analysis [1]. In alpine skiing, previous studies have investigated IMU sensor placement and performance analysis [2], turn detection [3], skiing-style classification [4], motion-quality quantification [5], [6], and recreational-skier feedback [7]. Multi-sensor inertial systems have also been used for joint and center-of-mass kinematics, training assessment, and skier attitude/level evaluation [8]–[11], while distributed IMU arrays have been explored for ski-mounted dynamic measurements [12]. However, boot-level systems provide limited information about upper-body and proximal lower-limb coordination, whereas full-body inertial motion capture requires more sensors and is less convenient for daily training. Similar compact sensor-integrated systems have also been explored in pipeline inspection and magnetic capsule localization [13], [14].

For beginner-oriented feedback, a lightweight system that captures key posture-related segments and reports interpretable motion problems is more practical than complete motion reconstruction or black-box skill scoring. This paper presents a five-node wireless IMU system for real-time skiing posture monitoring and motion quality feedback. The nodes are mounted on the left boot cuff, right boot cuff, left thigh, right thigh, and chest, allowing the system to capture lower-leg, thigh, and upper-body posture information with a compact hardware configuration. Static posture feedback is provided by a lower-limb alignment indicator and a backseat tendency indicator, while dynamic feedback is supported by reference-based curve comparison.

II. System Design

The proposed system consists of five wireless IMU nodes, an ESP32-based gateway, and desktop/mobile visualization terminals, as shown in Fig. 1. The nodes are mounted on the left boot cuff, right boot cuff, left thigh, right thigh, and chest to capture key lower-leg, proximal lower-limb, and upper-body posture information with a compact hardware configuration. Each sensing node integrates an ESP32-based controller, an MPU6050 IMU, a battery, and a compact enclosure. The MPU6050 provides three-axis acceleration and angular velocity measurements, from which roll and pitch angles are estimated [15]. In this five-node configuration, the boot-cuff nodes provide lower-leg alignment and forward-lean proxies, the thigh nodes capture proximal lower-limb posture, and the chest node reflects upper-body inclination, covering body segments related to lower-limb and trunk motion analysis in skiing [9], [11] while remaining lightweight and easy to deploy.

The sensing nodes transmit data to the ESP32 gateway through ESP-NOW. The gateway aggregates multi-node pack-

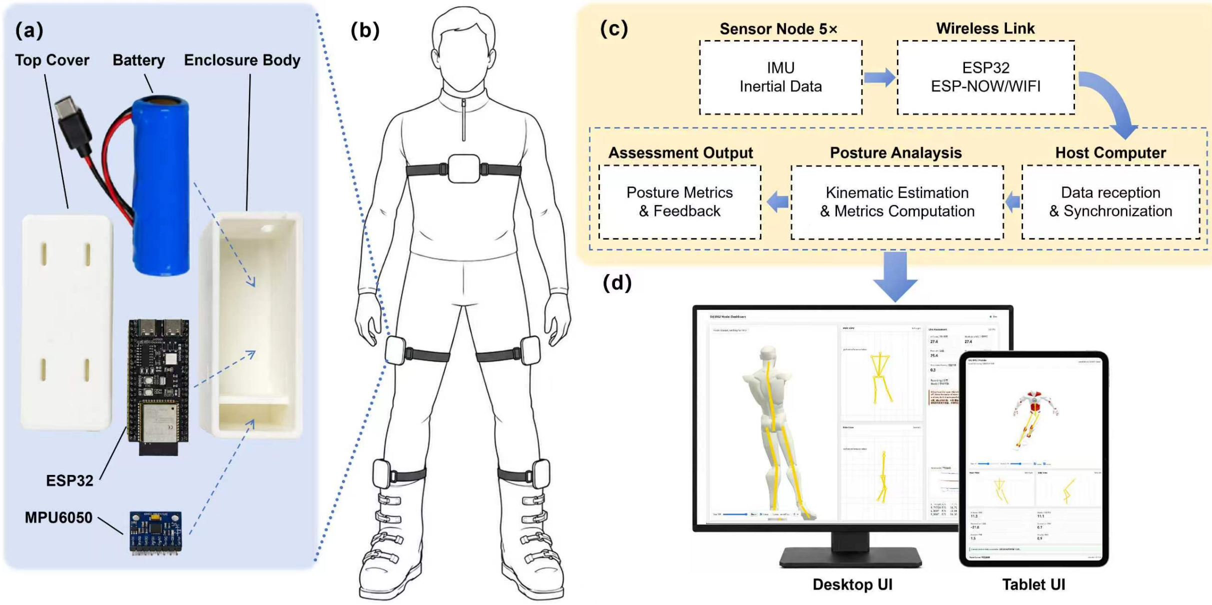

Fig. 1. Overview of the proposed multi-IMU skiing posture monitoring system. (a) Wearable IMU node. (b) Five-node wearing configuration on the left boot cuff, right boot cuff, left thigh, right thigh, and chest. (c) ESP-NOW sensing pipeline with an ESP32 gateway. (d) Computer/mobile interface for posture visualization, dynamic curves, local logging, and feedback.

ets and supports two downstream streaming modes: USB serial communication to a desktop dashboard for visualization, debugging, recording, and offline analysis, and Wi-Fi hotspot streaming to a mobile application for real-time posture display and local logging. This design allows the same sensing hardware to support both controlled experiments and practical field deployment.

III. Posture Reconstruction and Motion Quality Assessment

A. Angle Calibration and Key-Node Mapping

Before each trial, the skier stood upright in a neutral posture for 5 s. The mean roll and pitch angles during this period were used as node-specific references. The calibrated relative angle is calculated as

$$\tilde{\theta}_i(t) = s_i \left(\theta_i(t) - \theta_{i,0}\right), \tag{1}$$

where $\theta_i(t)$ is the real-time roll or pitch angle, $\theta_{i,0}$ is the neutral-posture reference, and $s_i \in \{-1, 1\}$ is a sign factor for mirrored sensor placement. This calibration reduces mounting offsets and short-term baseline bias, although long-term inertial drift may still remain.

Due to different mounting orientations, roll and pitch are mapped to frontal-plane and sagittal-plane posture angles with node-specific sign corrections. Superscripts f and s denote the corresponding frontal-plane and sagittal-plane angles. Since yaw is more susceptible to drift in the MPU6050-based configuration without an external heading reference, this prototype focuses on roll- and pitch-related posture cues, while yaw-related rotational errors are left for future work.

B. Static Posture Indicators

The static indicators describe two common beginner posture problems: frontal-plane lower-limb misalignment and sagittal-plane backseat tendency.

1) Lower-Limb Alignment Indicator: Let

$$\boldsymbol{\theta}^f_{LL}(t) = \left[\theta^f_{LT}(t), \theta^f_{RT}(t), \theta^f_{LB}(t), \theta^f_{RB}(t)\right]^T \tag{2}$$

denote the calibrated frontal-plane angles of the left thigh, right thigh, left boot cuff, and right boot cuff. The mean lower-limb frontal-plane angle is defined as

$$\bar{\theta}^f_{LL}(t) = \frac{1}{4}\mathbf{1}_4^T \boldsymbol{\theta}^f_{LL}(t), \tag{3}$$

and the lower-limb alignment indicator is defined as

$$I_{\mathrm{LLA}}(t) = \sqrt{\frac{1}{4}\left\|\boldsymbol{\theta}^f_{LL}(t) - \bar{\theta}^f_{LL}(t)\mathbf{1}_4\right\|_2^2}. \tag{4}$$

A low I_{LLA} indicates consistent frontal-plane alignment among the thigh and boot-cuff segments, while a high value suggests A-frame tendency, insufficient lower-leg parallelism, or other lower-limb alignment errors.

2) Backseat Tendency Indicator: The backseat tendency indicator is constructed from sagittal-plane chest, thigh, and boot-cuff angles. Let

$$\theta^s_B(t) = \frac{\theta^s_{LB}(t) + \theta^s_{RB}(t)}{2}, \; \theta^s_T(t) = \frac{\theta^s_{LT}(t) + \theta^s_{RT}(t)}{2}, \tag{5}$$

where $\theta^s_B(t)$ is the average boot-cuff forward-lean angle and $\theta^s_T(t)$ is the average thigh forward-swing angle. A simplified fore-aft posture score is defined as

$$x_{\mathrm{FA}}(t) = w_C \sin\theta^s_C(t) + w_B \sin\theta^s_B(t) - w_T \sin\theta^s_T(t). \tag{6}$$

Forward trunk inclination and boot-cuff forward lean increase x_{FA}, while thigh forward swing decreases it because it indicates that the hips move backward relative to the feet. The weights were set according to the normalized male body-segment mass proportions reported by de Leva [16], with $w_C = 0.54$, $w_T = 0.35$, and $w_B = 0.11$. The backseat tendency indicator is then defined as

$$I_{\mathrm{BS}}(t) = \max\left(0, x_{\mathrm{ref}} - x_{\mathrm{FA}}(t)\right), \tag{7}$$

where x_{ref} is the neutral-posture reference threshold. A larger I_{BS} indicates a stronger backseat-like tendency. In this work, I_{BS} is designed as an empirical fore-aft posture cue for beginner-oriented feedback, rather than as a biomechanically exact estimate of the center of mass or plantar pressure distribution.

C. Dynamic Motion Indicators and Reference-Based Comparison

For real-skiing analysis, each selected stable window is resampled to a normalized phase with M points, where $k = 1, \ldots, M$ is the phase index. Coach reference trials are used to construct reference curves, while rookie trials are used as comparison trials. Since skiing style, turn amplitude, and rhythm vary across individuals and segments, this comparison is intended for trial-level descriptive analysis rather than subject-level skill classification.

1) Upper-body engagement. The turn-related lower-limb curve is defined as

$$u(t) = \frac{\theta^f_{LB}(t) - \theta^f_{RB}(t)}{2}, \tag{8}$$

where $\theta^f_{LB}(t)$ and $\theta^f_{RB}(t)$ are the mapped frontal-plane boot-cuff angles. The upper-body engagement indicator is calculated as

$$I_{\mathrm{UBE}} = \frac{P_{95}\left(\theta^f_C\right) - P_5\left(\theta^f_C\right)}{P_{95}\left(u\right) - P_5\left(u\right) + \epsilon}, \tag{9}$$

where θ^f_C is the chest frontal-plane curve, $P_{95}(\cdot)$ and $P_5(\cdot)$ denote the 95th and 5th percentiles of a curve within the selected window, and ϵ is a small positive constant. A larger I_{UBE} indicates stronger lateral upper-body participation relative to lower-limb turning motion.

2) Turn-pattern deviation. For the coach reference trials, the mean turn-related curve is

$$\bar{u}_{\mathrm{coach}}(k) = \frac{1}{N_c}\sum_{j=1}^{N_c} u_j(k), \quad k = 1, \ldots, M, \tag{10}$$

where N_c is the number of coach reference trials. The turn-pattern deviation D_u is calculated as the root-mean-square difference between the test trial curve $u(k)$ and the coach reference curve $\bar{u}_{\mathrm{coach}}(k)$. A smaller D_u indicates a turn pattern closer to the coach reference pattern.

3) Lower-limb alignment deviation. Using the previously defined $I_{\mathrm{LLA}}(t)$, the coach reference curve is

$$\bar{I}_{\mathrm{LLA,coach}}(k) = \frac{1}{N_c}\sum_{j=1}^{N_c} I_{\mathrm{LLA},j}(k), \quad k = 1, \ldots, M. \tag{11}$$

The lower-limb alignment deviation D_{LLA} is calculated as the root-mean-square difference between the test trial curve $I_{\mathrm{LLA}}(k)$ and the coach reference curve $\bar{I}_{\mathrm{LLA,coach}}(k)$. A smaller D_{LLA} indicates a lower-limb coordination pattern closer to the coach reference pattern.

4) Backseat tendency. For dynamic trials, the window-averaged backseat tendency is used:

$$\bar{I}_{\mathrm{BS}} = \frac{1}{M}\sum_{k=1}^{M} I_{\mathrm{BS}}(k). \tag{12}$$

5) Other observed curve-level issues. Local problems such as left-right imbalance, abrupt transitions, A-frame-like misalignment, weak boot-cuff response, and weak thigh–boot-cuff coordination are inspected from representative curve segments rather than reported as primary numerical indicators.

IV. Experiments and Results

A. Controlled Angle Validation

To evaluate the angular measurement accuracy of the proposed IMU node, a video-based reference measurement was used for comparison. As shown in Fig. 2, an ArUco marker [17] was fixed to the same rigid housing as the IMU module, so that the marker and the IMU moved as a single rigid body. During each trial, the marker motion was recorded by a fixed camera, and the marker orientation was extracted frame by frame using computer vision. The obtained video-based angle was used as the reference angle for validating the IMU-estimated angle.

The video-based ArUco measurement was used as an approximate external angular reference rather than a high-precision motion-capture ground truth. The independent accuracy of the camera-based pose estimation was not quantified in this study, which is a limitation of the current validation.

Four controlled dynamic motion trials were conducted, including front sinusoidal, front triangular, lateral sinusoidal, and lateral triangular motions. To reduce the influence of starting and stopping segments, only the stable interval from 2.5 s to 17.5 s was used for quantitative analysis. After timestamp alignment, the IMU-estimated angle and the video-based reference angle were compared using MAE, RMSE, maximum error, and Bland-Altman agreement analysis.

Fig. 2 shows that the IMU-estimated angle follows the video-based reference angle closely in all four controlled motion trials. The results indicate that the IMU node can track both smooth continuous and piecewise linear angle variations. Across the four trials, the average MAE and RMSE were 0.6767° and 0.8119°, respectively, with the maximum error remaining within 3.1753°. The Bland-Altman bias values ranged from 0.2507° to 0.6778°, indicating only a small systematic overestimation. These results demonstrate that the proposed IMU node provides sufficient angular accuracy for posture-level skiing feedback.

B. Static Posture Selectivity Experiment

To evaluate whether the proposed posture indicators can selectively respond to different skiing posture problems, a static

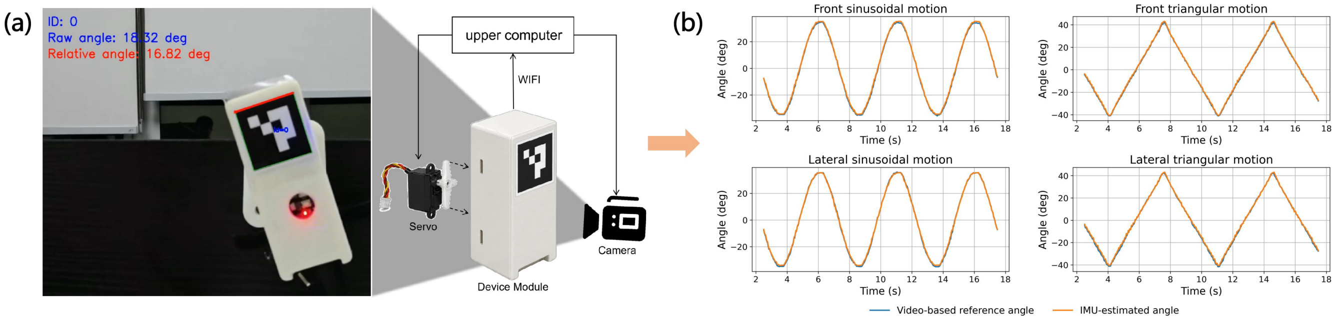

Fig. 2. Controlled angle validation of the proposed IMU node. (a) Video-based reference measurement setup with an ArUco marker attached to the same rigid housing as the IMU module. (b) Angle curve comparison between the IMU-estimated angle and the video-based reference angle under four controlled dynamic motions, including front sinusoidal, front triangular, lateral sinusoidal, and lateral triangular motions.

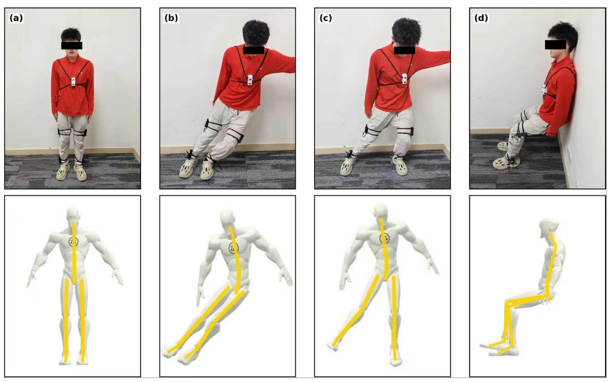

Fig. 3. Representative static posture cases and corresponding UI snapshots. The top row shows the four tested static postures: (a) neutral standing, (b) parallel edging stance, (c) A-frame posture, and (d) backseat posture. The bottom row shows the corresponding UI outputs of the proposed posture monitoring system.

posture experiment was conducted under four representative conditions: neutral standing, parallel edging stance, A-frame posture, and backseat posture. The neutral standing posture was used as the reference condition. The parallel edging stance represented a normal skiing posture with clear edging inclination, while the A-frame and backseat postures were used to activate the lower-limb alignment indicator I_{LLA} and the backseat tendency indicator I_{BS}, respectively. The A-frame posture was treated as a frontal-plane lower-limb misalignment pattern, which is conceptually related to dynamic knee valgus mechanics [18].

For each condition, the participant maintained the posture for several seconds, and a stable 7-s segment was selected for quantitative analysis. Representative photographs and corresponding UI snapshots are shown in Fig. 3. The lower-limb alignment indicator I_{LLA} was computed from the frontal-plane responses of the thigh and boot-cuff nodes, while the backseat tendency indicator I_{BS} was computed from the fore-aft posture score x_{FA} using chest, boot-cuff, and thigh sagittal-plane information. Fig. 4 summarizes the mean responses of the two indicators under the four posture conditions.

For I_{LLA}, neutral standing produced the lowest response, with a mean value of 1.05°. The parallel edging stance showed a moderate increase to 6.95°, which is reasonable because edging naturally involves body inclination and lower-limb engagement. The A-frame posture produced the largest response, reaching 18.37°, while the backseat posture yielded a smaller value of 3.84°. These results indicate that I_{LLA} mainly responds to frontal-plane lower-limb misalignment rather than sagittal-plane posture changes.

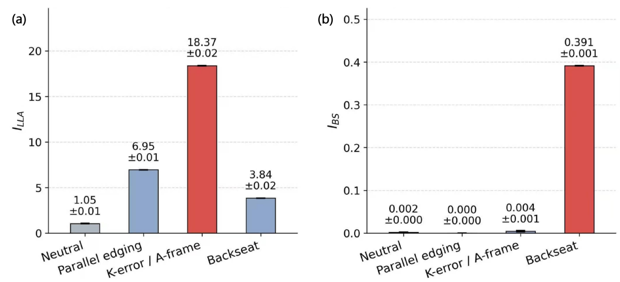

Fig. 4. Static posture indicator responses under four representative static postures. (a) Lower-limb alignment indicator I_{LLA}. (b) Backseat tendency indicator I_{BS}. Error bars and text labels indicate the 95% confidence interval half-width estimated from the selected stable segment.

For I_{BS}, neutral standing, parallel edging, and A-frame posture remained close to zero, with mean values of 0.0017, approximately 0, and 0.0044, respectively. In contrast, the backseat posture produced the largest response, reaching 0.3913. This confirms that I_{BS} selectively captures rearward fore-aft posture deviations and does not simply react to edging-related body inclination. Overall, the static posture experiment shows that the two proposed indicators provide complementary information: I_{LLA} is mainly sensitive to A-frame-like lower-limb misalignment, whereas I_{BS} is mainly sensitive to backseat-like posture.

C. Indoor Skiing Pilot Experiment

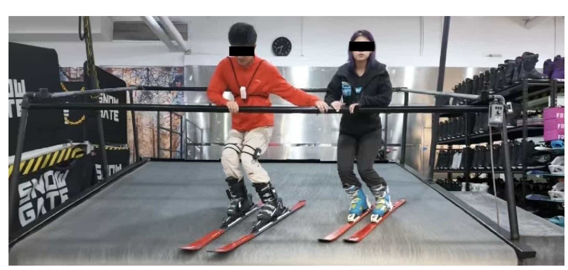

Fig. 5. Indoor skiing treadmill pilot experiment using the proposed five-node IMU system.

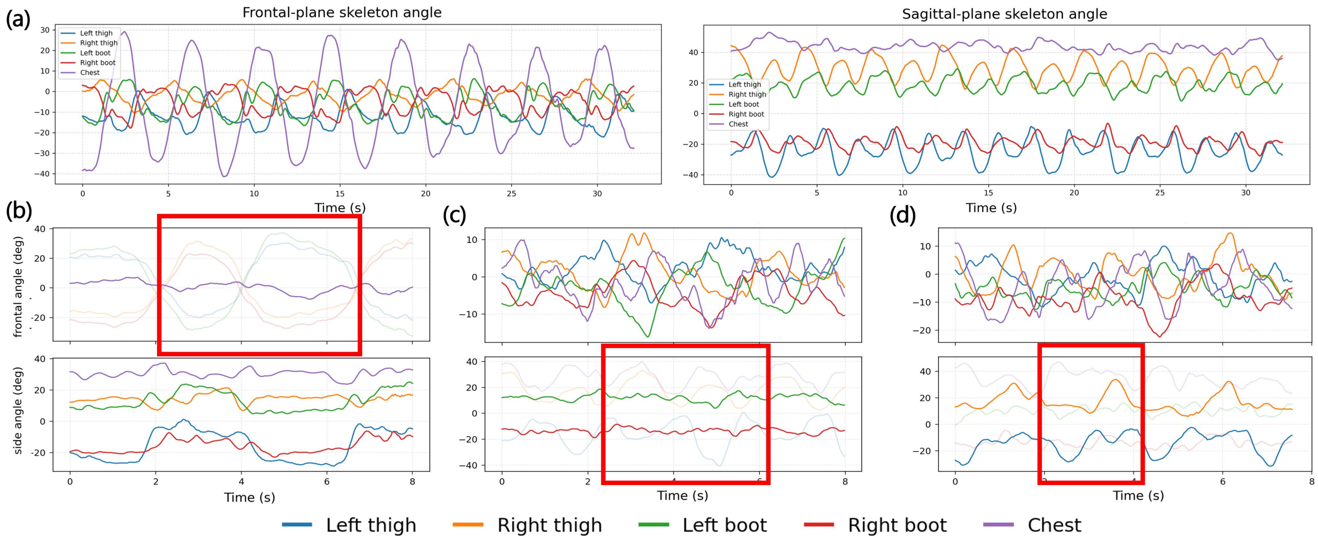

Fig. 6. Representative dynamic curves and typical rookie skiing problems. (a) Representative coach trial. (b) Rookie example with insufficient upper-body engagement. (c) Rookie example with weak boot-cuff response. (d) Rookie example with clear left-right asymmetry.

To further evaluate the feasibility of the proposed system under controlled indoor skiing conditions, a pilot experiment was conducted on an indoor skiing treadmill. As shown in Fig. 5, one coach participant and one rookie participant wore the proposed five-node IMU system and performed skiing motions on the treadmill with a similar turning rhythm. In total, seven coach trials and forty-one rookie trials were collected from selected stable skiing segments. Since the trials came from two participants rather than independent subject groups, they were used for trial-level descriptive comparison, and subject-level statistical significance testing was not performed.

After preprocessing and stable-window selection, the trials were analyzed using the proposed dynamic indicators, including upper-body engagement, turn-pattern deviation, lower-limb alignment deviation, and backseat tendency. Representative curve segments were also inspected to identify typical rookie skiing problems.

Fig. 6 shows one representative coach trial and three representative rookie problem cases. The coach trial shows a clear and repeated turning rhythm: the left and right boot-cuff curves vary periodically, the chest frontal-plane curve changes consistently with the turning pattern, and the sagittal-plane chest angle remains relatively stable, indicating coordinated lateral upper-body participation and stable forward-lean control. The thigh and boot-cuff curves also show a more organized temporal relationship. In contrast, Fig. 6(b) shows limited chest lateral variation despite clear lower-limb turning motion, indicating insufficient upper-body engagement. Fig. 6(c) shows weak boot-cuff amplitude changes that do not follow the skiing rhythm sufficiently, suggesting insufficient lower-leg involvement. Fig. 6(d) shows a weaker right-turn response than the corresponding left-turn response, indicating possible left-right asymmetry.

Fig. 7 shows the trial-level descriptive comparison. The coach trials showed higher upper-body engagement than the

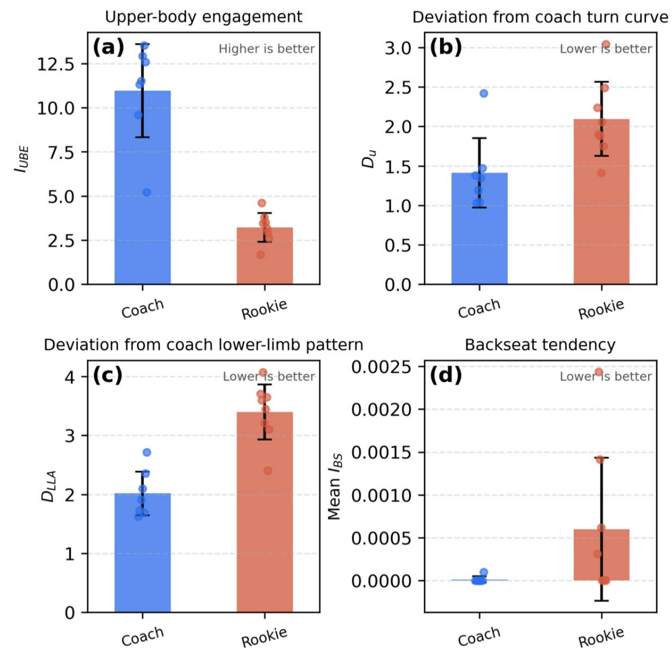

Fig. 7. Trial-level descriptive comparison between the coach participant and the rookie participant in the indoor skiing pilot experiment. (a) Upper-body engagement I_{UBE}. (b) Turn-pattern deviation D_u. (c) Lower-limb alignment deviation D_{LLA}. (d) Mean backseat tendency $\bar{I}_{\mathrm{BS}}$.

rookie trials, with mean I_{UBE} values of 10.9584 and 3.2226, respectively. The rookie trials also showed larger deviations from the coach reference pattern, with mean D_u increasing from 1.4121 to 2.0963 and mean D_{LLA} increasing from 2.0180 to 3.3992. These results indicate less consistent turning rhythm and lower-limb coordination in the selected rookie trial segments.

Since skilled skiers may naturally show larger lower-limb motion amplitudes during active edging, the absolute value of I_{LLA} is not directly interpreted as “higher is worse” in dynamic skiing. Instead, D_{LLA} is used to describe the deviation from the coach reference pattern. For backseat tendency, the coach trials showed a near-zero mean $\overline{I}_{\mathrm{BS}}$ of 0.000014, while the rookie trials showed a higher value of 0.000598. This result suggests that some rookie trial segments contained a stronger backseat-like tendency, while $\overline{I}_{\mathrm{BS}}$ remains a coarse sagittal-plane posture cue rather than an exact center-of-mass or pressure-distribution measurement.

Overall, the indoor skiing pilot experiment shows that the proposed five-node IMU system can capture interpretable motion differences in selected coach and rookie trial segments. Together with the representative curve examples, the results support the feasibility of the system for real-time skiing posture monitoring and preliminary motion feedback.

V. Conclusion

This paper presented a five-node wearable IMU system for real-time alpine skiing posture monitoring and interpretable motion feedback. The system uses five wireless nodes on key body segments and supports both desktop and mobile visualization. By combining static posture indicators and reference-based dynamic curve analysis, the proposed system provides key-segment feedback for common beginner skiing errors.

Controlled angle validation showed that the proposed IMU node achieved an average MAE of 0.6767° and RMSE of 0.8119°, indicating sufficient accuracy for posture-level feedback. Static posture experiments showed that the lower-limb alignment indicator responded most strongly to A-frame posture, while the backseat tendency indicator was mainly activated by backseat-like posture. These results demonstrate that the two indicators provide complementary and selective posture information.

The indoor skiing treadmill experiment further showed that the system can capture interpretable dynamic differences between coach and rookie trials. The rookie trials showed weaker upper-body participation, larger turn-pattern deviation, larger lower-limb alignment deviation, and slightly higher backseat tendency. Representative curve segments also revealed practical problems such as weak boot-cuff response, left-right asymmetry, and unstable transitions.

The current prototype is intended as a portable sensing and feedback platform rather than a replacement for expert coaching or full-body motion capture. The boot-cuff nodes provide useful lower-leg proxies but do not directly measure ski edge angle or ground reaction force, and the backseat tendency indicator is a posture cue rather than an exact center-of-mass or pressure-distribution measurement. Future work will include larger-scale skiing data collection, subject-specific calibration, improved turn segmentation, and more robust reference-curve construction with coach annotations.

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